TON_IoT

IoT/IIoT datasets designed to benchmark AI-based cybersecurity models

Based on: *Moustafa, N. (2021). A new distributed architecture for evaluating AI-based security systems at the edge: Network TON_IoT datasets. Sustainable Cities and Society.*

Presented by: Teresa Jency Bala | Khulna University | CSE Discipline

Industry 4.0 **IoT Security** **AI / ML** **UNSW Canberra**

MOTIVATION

Why Do We Need IoT-Specific Security Datasets?



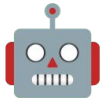
IoT Devices Are Everywhere

10B+ IoT devices globally generating network traffic and all potentially vulnerable to attack.



Old Datasets Are Obsolete

KDD99 and DARPA are among some older dataset which don't reflect modern IoT protocols, or new attack vectors like ransomware and MQTT-based intrusions.



AI Needs Real Labelled Data

Machine & Deep Learning models for intrusion detection must be trained and validated on datasets that mirror actual IoT network behavior not synthetic approximations.

INTRODUCTION

What are in the TON_IoT Datasets?

T - Telemetry (IoT/IIoT)

- 10+ sensor types: weather, Modbus, pH, soil moisture and more
- Protocol: MQTT via Mosquitto broker and Node-RED JavaScript
- Format: LOG and CSV files
- Captures normal sensor readings AND readings under active attack conditions
- 34 features, 4 per device type (temperature, humidity, status, location)
- Use case: device-level anomaly detection

O - OS Traces (Linux)

- Systems: Ubuntu 14 LTS and Ubuntu 18 TLS virtual machines
- Collection tool: atop; Linux system activity tracing monitor
- Three sub-tables: Disk activity | Memory usage | Process scheduling
- 34 features across 3 linked sub-tables, all linked via PID column
- Format: TXT and CSV
- Use case: Host-based Intrusion Detection (HIDS)

INTRODUCTION

What are in the TON_IoT Datasets?

O - OS Traces (Windows)

- Systems: Windows 7 and Windows 10 virtual machines
- Collection tool: Performance Monitor Tool (.blg format converted to CSV)
- Five subsystem groups: Processor | Process | Memory | Network | Disk
- 133 features across 5 subsystem categories
- Format: BLG (raw) and CSV (processed)
- Use case: Host-based Intrusion Detection (HIDS)

N - Network Traffic [OUR FOCUS]

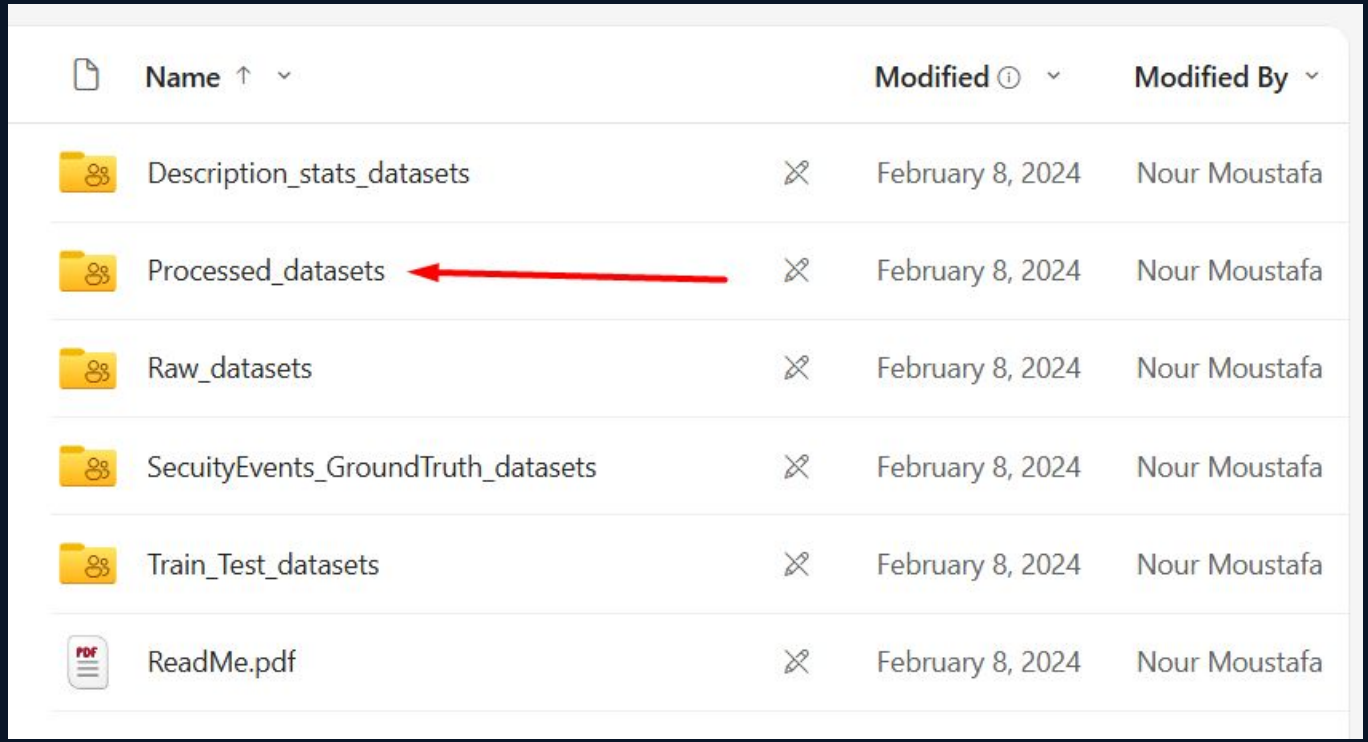
- ~8.54 million flow records from all testbed VMs, IoT devices, and cloud services
- Collection tool: ZEEK (Bro) framework processing raw PCAP packet captures
- 44 features: connection, protocol, DNS, HTTP, SSL, anomaly indicator groups
- Multi-class label: 9 attack types: DoS, DDoS, Backdoor, Injection, and more
- Use case: Network-based IDS (NIDS): ML/DL classification tasks

Dataset

The Ton_IoT dataset

[Dataset Link]

[Kaggle]



ARCHITECTURE

The Testbed: Edge - Fog - Cloud IoT Architecture

CLOUD LAYER (headquarter)

- Orchestrated Server (VMware NSX vCloud NFV)
- Cloud Analytics & Visualization Services
- Telemetry Logger via HTTPS
- Public HIVE MQTT Dashboard
- Public Vulnerable Websites (attack targets)

SDN/NFV Communication

FOG LAYER (local office)

- VMware NSX Data Center (vSphere Controller)
- SDN Management & NFV Data Cluster
- Middleware Server: FTP, DNS, Email, Kerberos, DHCP
- Security Onion (NIDS monitoring)
- Kali Linux VMs (attack launchers)
- Metasploitable3 & DVWA (vulnerable targets)
- Ostinato Traffic Generator

Internet Service / Physical Gateways

EDGE LAYER (physical layer)

- IoT/IIoT Devices: Weather sensors, Modbus, Smart Phones
- Host Systems: HP workstations, Dell laptops
- Physical Gateways: Routers & Switches
- Windows 7/10 VMs, Ubuntu 14/18 TLS VMs
- Node-Red JS (MQTT), Mosquitto MQTT Broker

NETWORK DATASET

TON_IoT Network Dataset - At a Glance

8.54M	44	9	21	2019
Total Flow Records	Network Features	Attack Categories	CSV	Collection Period

How the Network Data Was Collected

- The ZEEK (formerly BRO) network analysis framework processed raw PCAP packet captures into structured, labeled flow records (CSV).
- Attacks were launched by Kali Linux machines (IPs 192.168.159.30–39); ground truth labels were assigned based on attacker IPs + timestamps.
- Both a binary label (0 = normal, 1 = attack) and a multi-class 'type' label (e.g., 'ddos', 'backdoor') are included in each record.

FEATURES

The 44 Network Features: Categories Overview

Connection Identity — 5
ts, uid, src_ip, src_port, dst_ip, dst_port

Protocol & Service — 3
proto, service, conn_state

Flow Duration & Timing — 3
duration, missed_bytes, history

Byte Counts — 4
src_bytes, dst_bytes, src_ip_bytes, dst_ip_bytes

Packet Counts — 4
src_pkts, dst_pkts, src_ip_pkts, dst_ip_pkts

DNS Features — 7
dns_query, dns_qclass, dns_qtype, dns_rcode, dns_AA, dns_RD, dns_rejected

HTTP Features — 6
http_method, http_uri, http_version, http_status_code, http_request/response_body_len

SSL / Weird Features — 5
ssl_version, ssl_cipher, ssl_resumed, ssl_established, weird_name

Labels — 2
label (binary), type (multi-class attack name)

FEATURE DEEP DIVE

Time-Based & Flow-Based Features: The Core Analytics

Feature	Type	Description	Why It Matters for Classification
ts	Timestamp (Unix)	Start time of the network connection	Temporal attack patterns; DoS = tight burst windows; DDoS = distributed over time
duration	Float (seconds)	Total duration of the flow/connection	Short spikes → DoS/DDoS; Long connections → Backdoor, C&C
src_bytes	Integer	Bytes sent from source to destination	High src_bytes with small dst_bytes → data exfiltration or flooding
dst_bytes	Integer	Bytes sent from destination to source	Zero dst_bytes + high src_bytes → one-way flooding (DoS/DDoS)
src_pkts	Integer	Packets sent from source to destination	Packet rate: DoS floods packet counts; Scanning sends few packets
dst_pkts	Integer	Packets sent from destination to source	Zero dst_pkts → destination is not responding (blocked/overwhelmed)
src_ip_bytes	Integer	IP-level bytes from source (incl. headers)	Volume metric; higher than src_bytes by IP overhead signals network load
dst_ip_bytes	Integer	IP-level bytes from destination	Used with src_ip_bytes to compute byte ratio / traffic asymmetry
missed_bytes	Integer	Bytes from missed packets in reassembly	High missed_bytes → network congestion or packet-injection attacks
history	String flags	Connection history flags (S=SYN, F=FIN, etc.)	S0 (SYN only, no response) → SYN flood; OTH → various anomalies

FEATURE DEEP DIVE

Protocol, Service & Connection State Features

Feature	Values / Example	Description & Classification Relevance
proto	tcp / udp / icmp	Transport protocol. DoS/DDoS often exploits UDP & ICMP for amplification; TCP for SYN floods.
service	http / dns / ftp / smtp / ssh / ssl / dhcp / —	Application-layer service detected. Injection → HTTP/DNS; Scanning → multiple services; Backdoor → ssh/ftp.
conn_state	S0 / S1 / SF / REJ / RSTO / SH / OTH	Connection state: S0 = SYN only (flood); SF = normal close; REJ = rejected; RSTO = RST by originator.
src/dst ip	IPv4 addresses	Source/destination IPs. Note: high cardinality of src_ip → DDoS (many bots); single src_ip → DoS.
src_port / dst_port	0–65535	Port numbers. Known ports (80, 443, 22) = legitimate targets; high ephemeral ports = scanning/fuzzing.
dns_query	Domain string	DNS lookup domain. Malformed or repeated queries → DNS tunneling or exfiltration.
dns_rejected	Boolean	Whether DNS query was rejected. Mass rejections → DNS amplification attack pattern.

FEATURE DEEP DIVE

Protocol, Service & Connection State Features

Feature	Values / Example	Description & Classification Relevance
http_method	GET / POST / PUT	HTTP method. POST flood → HTTP DDoS; unusual PUT/DELETE → injection attempts.
http_status_code	200, 403, 500...	Server response code. Mass 500 errors → server overload from DoS; 200 flood → successful attack requests.
ssl_established	Boolean	Whether TLS session completed. Failed SSL handshakes → SSL exhaustion attacks.
weird_name	String	ZEEK-flagged anomaly name (e.g., 'bad_checksum'). Presence indicates protocol-level anomaly or manipulation.
label	0 (normal) / 1 (attack)	Binary classification target the primary output variable for intrusion detection.
type	Normal / ddos / dos / backdoor / injection / mitm/password/ransomware/scanning/xss	Multi-class target: 10 distinct classes (incl. normal). The primary output for multi-class classification.

ATTACK TYPES

The 9 Attack Types in TON_IoT Network Dataset

💥 DoS

Single-source flooding to exhaust victim resources

🌐 DDoS

Multi-source coordinated flooding via botnets

🚪 Backdoor

Hidden persistent access via Metasploit exploits

💉 Injection

SQL/command injection via web application flaws

🔍 Scanning

Network reconnaissance using Nmap / Nessus

👥 MITM

Man-in-the-Middle interception of network traffic

🔒 Ransomware

Encrypts victim files via SMB vulnerability exploit

📝 XSS

Cross-site scripting via web service injection

🔑 Password

Brute-force / credential stuffing attacks

ATTACK ANALYSIS

DoS & DDoS Attacks: The Volume-Based Threats

DoS: Denial of Service

Tool: UFONet toolkit + Python scripts

How it works:

Single attacker overwhelms target with high-volume traffic (SYN floods, HTTP floods, ICMP ping floods), depleting CPU/RAM/bandwidth.

Traffic signatures:

- S0 conn_state (SYN with no reply)
- Very high src_pkts, near-zero dst_pkts
- Short duration, massive byte counts
- Single src_ip, single dst_ip
- proto: tcp/udp/icmp

Dataset share: ~21% of attack records

DDoS: Distributed DoS

Tool: Coordinated scripts across multiple Kali VMs

How it works:

Hundreds of bot machines (often compromised IoT) simultaneously flood the target, making IP blocking ineffective the defining feature vs. DoS.

Traffic signatures:

- High cardinality of unique src_ip addresses
- Same dst_ip from many sources
- Gini coefficient → near 0 (uniform distribution)
- Synchronized burst patterns across temporal windows

Dataset share: ~71% of all labeled records (dominant class)

ATTACK ANALYSIS

Backdoor, Injection & Scanning Attacks

🚪 Backdoor

Tool: Metasploit Framework (bash persistence scripts)

Attacker gains persistent hidden access to a compromised host, maintaining a Command & Control (C&C) remains stealthy under normal traffic.

Network Signatures:

- Long-duration, low-bandwidth connections
- Persistent same src_ip–dst_ip pairs across time
- Unusual ports (e.g., 4444 Metasploit default)
- history: SF (full connection maintained)
- Encrypted tunnels → ssl_established = T

💉 Injection

Tool: Python scripts against DVWA / OWASP Shepherd

SQL injection, command injection, or XML injection through vulnerable web forms and APIs.

Network Signatures:

- HTTP POST with large http_request_body_len
- dns_query with encoded payloads
- http_status_code 500 (server errors)
- service = http, proto = tcp
- Repeated flows to dst_port 80/443/3306

🔍 Scanning

Tool: Nmap + Nessus vulnerability scanner (Kali)

Reconnaissance to discover live hosts, open ports, running services, and vulnerabilities. Produces highly characteristic traffic

Network Signatures:

- Large volume of short-duration flows (conn_state: S0/REJ)
- Many unique dst_ports from single src_ip
- Very low src_bytes, very low dst_bytes
- ICMP echo requests (proto = icmp)
- Systematic sequential port/IP sweeping pattern

ATTACK ANALYSIS

MITM, Ransomware, XSS & Password Attacks

👥 MITM

Attacker intercepts communications between two parties e.g., ARP spoofing on LAN causes IoT sensor data to be rerouted through attacker.

Key signals: Bi-directional flows; multiple connections per IP pair;
ARP-related protocol anomalies; ssl_established changes

🔒 Ransomware

Metasploit exploits Windows SMB vulnerability (EternalBlue). Encrypts victim files and demands payment. Very destructive targets Windows VMs monitoring IoT.

Key signals: SMB traffic surge (port 445); Large file transfers; Unusual write operations; Windows-specific process events

📝 XSS

Cross-Site Scripting injected into vulnerable web apps. Malicious JavaScript executes in other users' browsers steals credentials or hijacks sessions.

Key signals: HTTP POST to web forms; script-like strings in http_uri; Repeated hits to same page from same IP; http_status_code 200

🔑 Password

Brute-force or credential stuffing against SSH, FTP, HTTP login forms and MQTT broker authentication. Uses automated tools from Kali Linux.

Key signals: Rapid repeated connections to same dst_port; Many short flows (conn_state = S1/REJ); High dst_pkts from server (error responses)

CLASS DISTRIBUTION

Class Distribution: Understanding the Imbalance



KEY INSIGHTS

Scanning #1 32.0%
Largest class: DDoS. Nmap/Nessus recon dominates the full dataset.

DDoS #2 27.6%
Second largest. Combined Scanning + DDoS = ~60% of all records.

Normal 3.6%
Only 796K benign flows out of 22.3M. Extreme minority

MITM 0.005%
Just 1,052 records. The rarest class: a true few-shot learning problem.

Metric rule
Always report Macro F1, MCC, per-class Precision/Recall: never overall accuracy.

LABELING STRATEGY

How Ground Truth Was Created: Labeling Methodology

01

Attacker IP Allocation

Kali Linux attack machines were assigned a fixed IP range: 192.168.159.30 – 192.168.159.39. This created a reliable, pre-known set of 'attacker' source IPs.

02

Parallel Attack Execution

Attacks were launched simultaneously while legitimate IoT operations continued realistic co-existence of benign and malicious traffic, not separate captures.

03

Timestamp Logging

The SecurityEvents_GroundTruth folder records exact start/end timestamps of every attack session, enabling precise correlation with network flow records.

04

IP + Timestamp Tagging

Each ZEEK flow record is labeled: if src_ip ∈ attacker range AND ts falls within attack window → label=1, type=<attack_name>. Otherwise → label=0, type='normal'.

05

Dual Label Structure

Two label columns: 'label' (binary: 0/1) supports Binary Classification. 'type' (multi-class: normal/ddos/dos/backdoor/...) supports Multi-Class Classification.

CLASSIFICATION TASK

How to Use TON_IoT for Classification

⚖ BINARY CLASSIFICATION

Target variable:	label column (0 = normal, 1 = attack)
Question:	Is this network flow benign or malicious?
ML Models:	Logistic Regression, SVM, Random Forest, XGBoost, MLP, CNN-LSTM
Evaluation:	Accuracy, Precision, Recall, F1-score, AUC-ROC, MCC
Key Challenge:	Class imbalance: only ~7.5% normal traffic. Must use stratified split + class weights.
Best Use:	First-level IDS alert system

🎯 MULTI-CLASS CLASSIFICATION

Target variable:	type column (10 classes: normal + 9 attacks)
Question:	What type of attack is this flow (if any)?
ML Models:	Random Forest, XGBoost, GBM, Deep Neural Network, GNN
Evaluation:	Macro F1, Weighted F1, Confusion Matrix, Cohen's Kappa, MCC per class
Key Challenge:	Severe class imbalance + rare classes (XSS: 5126, MITM: 3862 records). Focal Loss or SMOTE needed.
Best Use:	Advanced threat intelligence & attack attribution

RESEARCH DIRECTIONS

What Can Be Done With TON_IoT (network)

CLASSICAL & Ensemble ML

Random Forest
fast, interpretable baseline

XGBoost / LightGBM
Top performer on tabular data

AdaBoost, Bagging, Stacking

Gradient Boosting
Handles imbalance well with class weights

SVM
Effective in lower dimensions

Naive Bayes
weak on correlated features; useful for speed

k-NN / Decision Tree
Good for prototyping; interpretable; overfits rare classes

DEEP LEARNING

MLP / Deep DNN
learns non-linear traffic boundaries

Autoencoder
Unsupervised: train on normal only; flag reconstruction error

Bi-LSTM, Bi-GRU

CNN (1D)
Treats feature vector as sequence; captures local patterns

CNN-LSTM Hybrid
Spatial + temporal jointly

LSTM / GRU
Temporal dynamics: burst detection, slow-drain attacks

Transformer / BERT
Attention over flow sequences; emerging in NIDS research

RESEARCH DIRECTIONS

What Can Be Done With TON_IoT (network)

GRAPH / GNN

GCN / GraphSAGE

Model flows as nodes; neighbourhood aggregation

GAT / GATv2 / TGNN

Dynamic attention on attacker-victim edges

E-GraphSAGE

Edge-feature aware GNN; captures protocol-level relations

FRONTIER

Federated Learning

Local training per IoT node; no raw data shared; homomorphic aggregation

GAN / CTGAN

Synthesise realistic minority attack flows (MITM: 3.8K, XSS: 5.1K) to fix class imbalance

LLM (Fine-tuned)

Flows → text prompts; QLoRA fine-tune LLaMA; RAG for zero-shot novel attack detection

Adversarial ML

FGSM / PGD perturbations on flow features; craft evasion attacks; harden IDS

Continual Learning

Concept drift over 27-day window; elastic weight consolidation

Also can try Hybrid approached

Other IoT Network Security Datasets: For Reference

These datasets exist and are used in the community

UNSW-NB15 2015	Size: 2.5M records, 49 features Attacks: 9 types	*Same lab (UNSW); no IoT sensors; strong baseline*
Bot-IoT 2018	Size: 72M records, 46 features Attacks: 5 types	*IoT botnet-specific; DDoS/DoS/Theft; from UNSW*
CICIDS2017 2017	Size: 2.8M records, 78 features Attacks: 14 types	*Canadian Institute for Cybersecurity; no IoT sensors*
Edge-IIoTset 2022	Size: 20.9M records, 61 features Attacks: 14 types	*Covers MQTT/CoAP/Zigbee IoT protocols; newer*
IoT-23 2020	Size: 20 captures, pcap Attacks: 20 malware types	*Real Raspberry Pi IoT device; unlabeled pcap format*

What Can You Build With TON_IoT?

Intrusion Detection Systems (IDS)

- Binary IDS: classify traffic as benign vs. malicious
- Multi-class IDS: identify exact attack type (9 classes)
- Real-time streaming IDS using sliding time windows
- Federated IDS across distributed IoT nodes

Feature Engineering Research

- Entropy-based features for DoS vs. DDoS separation
- Temporal aggregation: rolling window statistics
- Graph-level features from flow-to-flow topology
- Feature selection: SHAP, Boruta, Mutual Information

Class Imbalance Handling

- SMOTE / ADASYN oversampling for minority attacks
- Focal Loss training to penalise rare-class errors
- Cost-sensitive learning & class-weight tuning
- One-class classification for anomaly detection

What Can You Build With TON_IoT?

Advanced ML / DL Models

- Graph Neural Networks on flow-graph topology
- Transformer / Attention models on flow sequences
- Autoencoder-based unsupervised anomaly detection
- Adversarially robust models (PGD, FGSM hardening)

Explainability & Interpretability

- SHAP values: which features drive each attack type?
- GNNExplainer: which sub-graphs signal an attack?
- Attention weight visualisation across flow sequences
- Decision boundary analysis for rare attack classes

Cross-Dataset & Generalisation

- Transfer learning: train on TON_IoT, test on Bot-IoT
- Zero-shot detection of unseen attack types
- Domain adaptation across different IoT environments
- Benchmark comparison: TON_IoT vs. UNSW-NB15 vs. Edge-IIoTset

What We Covered Today

Motivation	Why IoT traffic needs its own security datasets
Where was TON_IoT made	Dr Nour Moustafa, UNSW Canberra, 2019–2021 | Edge–Fog–Cloud testbed
4 Data Sources	Telemetry (T), OS (O), Network (N) — real devices, real attacks
Network Dataset	8.54M flows | 44 features | ZEEK/Bro captured | binary + multi-class labels

What We Covered Today

44 Features	Timestamps, IPs, ports, protocol, bytes, packets, DNS, HTTP, SSL, conn_state
Time-Based	ts, duration, src_bytes/pkts, missed_bytes, history: each carrying attack signal
9 Attack Types	DoS, DDoS, Backdoor, Injection, Scanning, MITM, Ransomware, XSS, Password
Research Scopes	IDS, feature engineering, GNN,, adversarial robustness, transfer learning

📂 *Dataset: https://research.unsw.edu.au/projects/toniot-datasets | Cite: Moustafa, N. (2021). Sustainable Cities and Society.*



THANK YOU

Feel Free to ask Questions